

Iteration 1 (Bridget Karungi)

$$y = mx_1 + m_2x_2 + b$$

$$m = [-1, 2]$$

$$b = [1, 1]$$

Data points

$$x = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix}$$

$$y = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$$

① Predicted values

$$\hat{y} = X \cdot m + b$$

$$= \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1+6 \\ -4+20 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 16 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \underline{\underline{\begin{pmatrix} 6 \\ 17 \end{pmatrix}}}$$

$$\frac{\text{Gradient}}{d_j/dm} = \frac{\text{MSE}}{\frac{1}{n} \sum_{i=1}^n (y - \hat{y})^2}$$

② Error

$$E = y - \hat{y}$$

$$= \begin{pmatrix} 5 \\ 6 \end{pmatrix} - \begin{pmatrix} 6 \\ 17 \end{pmatrix} = \underline{\underline{\begin{pmatrix} -1 \\ -11 \end{pmatrix}}}$$

$$\begin{aligned}
 \textcircled{3} \text{ MSE} &= \frac{1}{n} \sum_{i=1}^n (y - \hat{y})^2 \\
 &= \frac{1}{2} \begin{pmatrix} 1 \\ 11 \end{pmatrix}^2 \\
 &= \frac{1}{2} (1 + 11^2) \\
 &= \frac{1}{2} (1 + 121) \\
 &= \frac{1}{2} (122) \\
 &= \underline{\underline{61}}
 \end{aligned}$$

$\textcircled{4}$ Gradient ^{with} descent respect to m	Gradient with respect to b
$ \begin{aligned} \Delta_m &= -\frac{2}{n} X^T (y - \hat{y}) \\ &= -\frac{2}{2} \begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix} \begin{pmatrix} -1 \\ -11 \end{pmatrix} \\ &= -1 \begin{pmatrix} -1 - 44 \\ -3 - 110 \end{pmatrix} \\ &= -1 \begin{pmatrix} -45 \\ -113 \end{pmatrix} \\ &= \underline{\underline{\begin{pmatrix} 45 \\ 113 \end{pmatrix}}} \end{aligned} $	$ \begin{aligned} \Delta_b &= -\frac{2}{n} \sum_{i=1}^n (y - \hat{y}) \\ &= -\frac{2}{2} \begin{pmatrix} -1 \\ -11 \end{pmatrix} \\ &= \underline{\underline{\begin{pmatrix} 1 \\ 11 \end{pmatrix}}} \end{aligned} $

⑤ Update

learning rate (α) = 0.01

Update m

$$m_{\text{new}} = m - \alpha \Delta_m$$

$$= \begin{pmatrix} -1 \\ 2 \end{pmatrix} - 0.01 \begin{pmatrix} 45 \\ 113 \end{pmatrix}$$

$$= \begin{pmatrix} -1 - 0.45 \\ 2 - 1.13 \end{pmatrix}$$

$$= \begin{pmatrix} -1.45 \\ 0.87 \end{pmatrix}$$

Update b

$$b_{\text{new}} = b - \alpha \Delta_b$$

$$= \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 0.01 \begin{pmatrix} 1 \\ 11 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 0.01 \\ 0.11 \end{pmatrix}$$

$$= \begin{pmatrix} 0.99 \\ 0.89 \end{pmatrix}$$

Iterate

Iteration 2 (Armstrong Hirwa)

1. Prediction

$$\begin{aligned}\hat{y} &= X \cdot m_{\text{new}} + b_{\text{new}} \\ &= \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} \\ &= \begin{bmatrix} -1.45 + 2.61 \\ -5.8 + 8.7 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} \\ &= \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix}\end{aligned}$$

2. Error

$$\text{Error} = y - \hat{y} = \begin{bmatrix} 5 \\ 6 \end{bmatrix} - \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix} = \begin{bmatrix} 2.85 \\ 2.21 \end{bmatrix}$$

3. Mean Squared Error (MSE) = $\frac{1}{n} \sum_{i=1}^n (y - \hat{y})^2$

$$\begin{aligned}&= \frac{1}{2} \begin{bmatrix} 2.85 \\ 2.21 \end{bmatrix}^2 \\ &= \frac{1}{2} (13.0066) = \boxed{6.5033}\end{aligned}$$

4. Gradient for m and b

$$\begin{aligned}\frac{dj}{dm} &= \frac{-2}{n} X^T (y - \hat{y}) \\ &= \frac{-2}{2} \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 2.85 \\ 2.21 \end{bmatrix} \\ &= - \begin{bmatrix} 2.85 + 2.85 \\ 8.55 + 2.21 \end{bmatrix} = \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix}\end{aligned}$$

$$\frac{dj}{db} = \frac{-2}{n} \sum_{i=1}^n (y - \hat{y}) = \frac{-2}{2} \begin{bmatrix} 2.85 \\ 2.21 \end{bmatrix} = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

5. Update.

$$\begin{aligned}m_{\text{new}} &= m_{\text{old}} - \alpha \frac{dj}{dm} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - 0.01 \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix} \\ &= \begin{bmatrix} -1.45 + 0.1169 \\ 0.87 + 0.3065 \end{bmatrix} = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix}\end{aligned}$$

$$\begin{aligned}b_{\text{new}} &= b_{\text{old}} - \alpha \frac{dj}{db} = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} - 0.01 \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix} \\ &= \begin{bmatrix} 0.99 + 0.0285 \\ 0.89 + 0.0221 \end{bmatrix} = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}\end{aligned}$$

Iteration 3 (Naïve)

1. Predicted values

$$\hat{y}'' = X \cdot m'_{\text{new}} + b'_{\text{new}}$$

$$= \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} \begin{pmatrix} -1.3331 \\ 1.1765 \end{pmatrix} + \begin{pmatrix} 1.0185 \\ 0.9121 \end{pmatrix}$$

$$= \begin{pmatrix} -1.3331 + 3 \cdot 5295 \\ -5 \cdot 3324 + 11 \cdot 765 \end{pmatrix} + \begin{pmatrix} 1.0185 \\ 0.9121 \end{pmatrix}$$

$$= \begin{pmatrix} 3.2149 \\ 7.3447 \end{pmatrix}$$

2. Error

$$E = y - \hat{y}''$$

$$= \begin{pmatrix} 5 \\ 6 \end{pmatrix} - \begin{pmatrix} 3.2149 \\ 7.3447 \end{pmatrix}$$

$$= \begin{pmatrix} 1.7851 \\ -1.3447 \end{pmatrix}$$

Gradient with respect to m'

$$\Delta m = -\frac{2}{n} X^T (y - \hat{y})$$

$$= -\frac{2}{2} \begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix} \begin{pmatrix} 1.7851 \\ -1.3447 \end{pmatrix}$$

$$= \begin{pmatrix} -1.7851 + 5.3788 \\ -5.3553 + 13.447 \end{pmatrix}$$

$$= \begin{pmatrix} 3.5937 \\ \underline{\underline{8.0917}} \end{pmatrix}$$

Gradient with respect to b'

$$\Delta b = -\frac{2}{n} (y - \hat{y})$$

$$= -\frac{2}{2} \begin{pmatrix} 1.7851 \\ -1.3447 \end{pmatrix}$$

$$= \begin{pmatrix} -1.7851 \\ \underline{\underline{1.3447}} \end{pmatrix}$$

Update m

$$m''_{\text{new}} = m - \alpha \Delta m$$

$$= \begin{pmatrix} -1.3331 \\ 1.1765 \end{pmatrix} - 0.01 \begin{pmatrix} 3.5937 \\ 3.5937 \end{pmatrix}$$

$$= \begin{pmatrix} -1.3331 - 0.035937 \\ 1.1765 - 0.035937 \end{pmatrix}$$

$$= \begin{pmatrix} -1.369037 \\ \underline{\underline{1.095583}} \end{pmatrix}$$

Update h

$$h''_{\text{new}} = h - \alpha \Delta h$$

$$h''_{\text{new}} = \begin{pmatrix} 1.0185 \\ 0.9121 \end{pmatrix} - 0.01 \begin{pmatrix} -1.7851 \\ 1.3447 \end{pmatrix}$$

$$= \begin{pmatrix} 1.0185 + 0.017851 \\ 0.912 - 0.013447 \end{pmatrix}$$

$$= \begin{pmatrix} 1.036351 \\ \underline{\underline{0.898653}} \end{pmatrix}$$

Hernton 4 (Normin)

1. Predicted values

$$\hat{y}''' = X \cdot m''_{\text{new}} + b''_{\text{new}} \\ = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} \begin{pmatrix} -1.369037 \\ 1.095583 \end{pmatrix} + \begin{pmatrix} 1.036351 \\ 0.898653 \end{pmatrix}$$

$$= \begin{pmatrix} 1.917712 \\ 5.479682 \end{pmatrix} \begin{pmatrix} 1.036351 \\ 0.898653 \end{pmatrix} \\ = \begin{pmatrix} 2.954063 \\ 6.378335 \end{pmatrix}$$

2. Error

$$E = y - \hat{y}''' \\ = \begin{pmatrix} 5 \\ 6 \end{pmatrix} - \begin{pmatrix} 2.954063 \\ 6.378335 \end{pmatrix} \\ = \begin{pmatrix} 2.045937 \\ -0.378335 \end{pmatrix}$$



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Update b

$$b_{\text{new}} = b = \alpha \Delta b$$

$$b = \begin{pmatrix} 1.036351 \\ 0.898653 \end{pmatrix}$$

$$b_{\text{new}} = \begin{pmatrix} 1.036351 \\ 0.898653 \end{pmatrix} - 0.01 \begin{pmatrix} -2.045937 \\ 0.378335 \end{pmatrix}$$

$$b_{\text{new}} = \begin{pmatrix} 1.036351 + 0.020459 \\ 0.898653 + -0.003783 \end{pmatrix}$$

$$b_{\text{new}} = \begin{pmatrix} 1.056810 \\ 0.894870 \end{pmatrix}$$

$$\Delta_m = \begin{pmatrix} -0.532597 \\ -2.354461 \end{pmatrix}$$

$$\Delta_b = \begin{pmatrix} -2.045937 \\ 0.378335 \end{pmatrix}$$



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$$m_{new} = \begin{pmatrix} -1.363711 \\ 1.119128 \end{pmatrix}$$

$$b_{new} = \begin{pmatrix} 1.056810 \\ 0.894870 \end{pmatrix}$$